

ML@B NMEP Homework 1: Math

Fall 2026

The goal of this homework is to build comfort with the mathematics that will appear repeatedly throughout machine learning. More important than getting through computations quickly is being able to reason about what mathematical objects mean, what shapes they have, and how different operations fit together.

When working with vectors and matrices, we strongly recommend writing down the shape of every intermediate quantity.

1. Vectors, Matrices, and Shapes

Suppose vectors are column vectors unless otherwise specified.

(a) Let

$$x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 2 & 1 \end{bmatrix}.$$

Compute $x^\top y$, $xy^\top$, Ax , and AB , and give the shape of each result. Which of $x^\top A$, Bx , $A^\top x$, and BA are well-defined?

(b) For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, express $(AB)_{ij}$ using a row of A and a column of B .

Now, count one scalar addition or multiplication as one FLOP (floating-point operation). To leading order, how many FLOPs are required to compute AB directly?

(c) A batch of B datapoints, each with d features, is stored as $X \in \mathbb{R}^{B \times d}$. We apply the same transformation to every datapoint and obtain $H = XW \in \mathbb{R}^{B \times h}$. Determine the shape of W and explain what the i th row of H represents.

Now suppose the data instead have shape $X \in \mathbb{R}^{B \times N \times d}$ and the same W is applied independently to each of the BN vectors in $\mathbb{R}^d$. Give the output shape and describe how the same computation could be performed using a single ordinary matrix multiplication by reshaping X before and after the multiplication.

(d) Suppose

$$X \in \mathbb{R}^{32 \times 128}, \quad W_1 \in \mathbb{R}^{128 \times 64}, \quad W_2 \in \mathbb{R}^{64 \times 10}.$$

Without performing any numerical computation, determine the shapes of XW_1 , XW_1W_2 , W_1W_2 , and $X(W_1W_2)$. Which two of these computations must give the same matrix, and why?

2. Splitting a Matrix Computation

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, with $n = n_1 + n_2$. Suppose

$$A = \begin{bmatrix} A_1 & A_2 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix},$$

where the blocks are split along the shared dimension.

- (a) Determine the shapes of A_1, A_2, B_1, B_2 , and express AB entirely in terms of these four matrices. Hence describe how two workers, each holding one corresponding pair (A_i, B_i) , could independently perform most of the computation of AB .

- (b) Suppose instead that

$$B = \begin{bmatrix} B_1 & B_2 \end{bmatrix}, \quad B_1 \in \mathbb{R}^{n \times p_1}, \quad B_2 \in \mathbb{R}^{n \times p_2}, \quad p_1 + p_2 = p.$$

If two workers compute AB_1 and AB_2 independently, how are their results combined to recover AB ? Why does this split require a different combination operation from part (a)?

3. Gradients, Jacobians, and the Chain Rule

For a scalar valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ taking $x \in \mathbb{R}^n$, its gradient $\nabla_x f \in \mathbb{R}^n$ is the vector of partial derivatives, as introduced in lecture. However, vector output functions also exist (consider a linear transformation!). It is important *not* to conflate the partial derivatives of scalar valued functions and vector valued functions, so always remind yourself of what type of function you are dealing with. For a vector valued function $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$, its Jacobian, the matrix of partial derivatives $J_g(x) \in \mathbb{R}^{m \times n}$ is defined by

$$[J_g(x)]_{ij} = \frac{\partial g_i}{\partial x_j}.$$

- (a) For fixed $a \in \mathbb{R}^n$, let

$$f(x) = (a^\top x)^2 + \frac{1}{2} x^\top x.$$

Compute $\nabla_x f$ and verify that the shape of your answer is consistent with the definition of a gradient.

- (b) Let

$$g(x_1, x_2) = \begin{bmatrix} x_1^2 + x_2 \\ x_1 x_2 \\ e^{x_1} + x_2^2 \end{bmatrix}.$$

Compute $J_g(x)$.

- (c) Let $g(x) = Wx + b$, where $W \in \mathbb{R}^{m \times n}$. Compute $J_g(x)$.

Now suppose $h(x) = W_2(W_1x + b_1) + b_2$. Determine $J_h(x)$ in terms of W_1 and W_2 , and explain how its shape follows from the shapes of the two transformations.

- (d) Let $z = Wx + b$ and

$$L(x) = \frac{1}{2} \|z - y\|_2^2.$$

Compute $\nabla_x L$. Your derivation should make clear where the transpose of W comes from.

4. Berkeley Class Sizes

A university has n classes with sizes $s_1, \dots, s_n$. Define

$$N = \sum_{i=1}^n s_i, \quad \mu = \frac{1}{n} \sum_{i=1}^n s_i = \frac{N}{n}.$$

You do not know the s_i . Instead, you choose uniformly at random from the N students leaving these classes and ask the selected student how large their class was (you can't ask the student which class they are in or any other information). Let I be the class of the selected student and let $S = s_I$ be their response.

- (a) Find $\Pr(I = i)$ and $\mathbb{E}[S]$. Explain why simply averaging many responses S does not, in general, estimate μ correctly.

- (b) Find

$$\mathbb{E} \left[\frac{1}{S} \right].$$

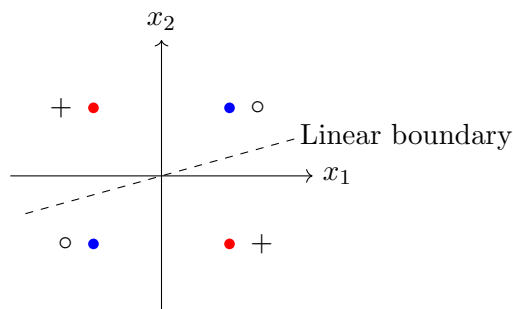
- (c) Suppose you independently sample k students and observe $S_1, \dots, S_k$. Construct an unbiased estimator of $1/\mu$. (For an unknown parameter θ which we are trying to estimate, the estimator $\hat{\theta}$ has bias $\mathbb{E}[\hat{\theta}] - \theta$, so an unbiased estimator of $\frac{1}{\mu}$ should have an expected value equal to $\frac{1}{\mu}$.)

- (d) Use your answer above to construct a natural estimator $\hat{\mu}$ of the average class size. Identify the familiar type of mean represented by $\hat{\mu}$.

Is $\hat{\mu}$ itself necessarily unbiased? Please explain to the best of your ability.

5. Linear Neural Networks

In their seminal 1969 book *Perceptrons*, MIT researchers Minsky and Papert showed that linear classifiers could not perfectly classify data according to the XOR operation. Naturally, they claimed, a model that learns a line should not be able to discriminate data which are nonlinearly correlated.

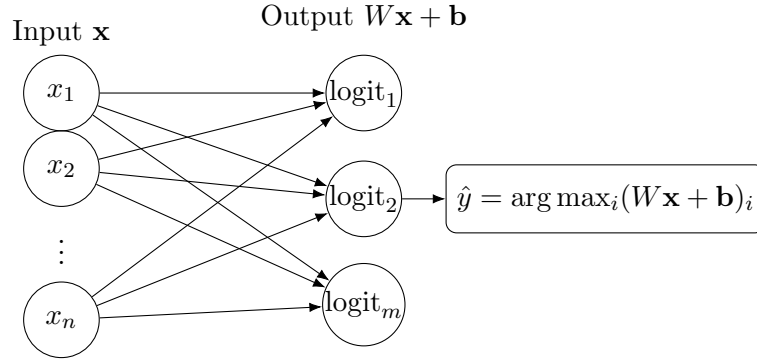


XOR data distribution centered at the origin.

Later work rectified this by imposing a **nonlinear activation function** through an intermediary **hidden layer**, allowing for nonlinear decision boundaries. In this problem, we will explore first why a linear classifier learns linear decision boundaries and second, why a hidden layer without nonlinearity is a redundancy.

- (a) A linear classifier $f_{\theta}(\mathbf{x})$ is a single layer neural network which takes a datapoint $\mathbf{x} \in \mathbb{R}^{n \times 1}$ and outputs a prediction $\hat{y} \in C$, where $C = \mathbb{R}^m$ is the set of classes and $y \in C$ is the true classification of $\mathbf{x}$. *Hint: n is the number of input features, m is the number of output classes*

$$\hat{y} = f_{\theta}(\mathbf{x}) = \arg \max_{i \in C} (W\mathbf{x} + \mathbf{b})_i \quad W \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^{m \times 1}$$



In other words, we predict the class for which the associated entry in $W\mathbf{x} + \mathbf{b}$, which we call the **logits** of this network, is greatest. Geometrically, we can think of this as partitioning the input space into m regions, where any two regions are separated by a set of inputs for which the logits associated with each region are equal, i.e. the **decision boundary**.

Show that each decision boundary is a hyperplane and, using this, informally explain why a linear classifier would not be able to perfectly classify a nonlinear data distribution.

Hint: A hyperplane in a Euclidean vector space is defined as a set $\{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T \mathbf{x} = b\}$ for some $\mathbf{a} \in \mathbb{R}^n, b \in \mathbb{R}$.

- (b) Show that adding more layers to this network is redundant.